torque $\tau_B$ at constant angular velocity, then the work done by external torque $(\vec{\tau}_{ext})$ for this small angular displacement is given by

$$dW = |\vec{\tau}_{ext}| d\theta$$

The bar magnet has to be moved at constant angular velocity, which implies that $|\vec{\tau}_B| = |\vec{\tau}_{ext}|$

$$dW = p_m B \sin\theta \, d\theta$$

Total work done in rotating the dipole from $\theta'$ to $\theta$ is

$$W = \int_{\theta'}^{\theta} \tau d\theta = \int_{\theta'}^{\theta} p_m B \sin\theta d\theta = p_m B \left[-\cos\theta\right]_{\theta'}^{\theta}$$

$$W = -p_m B(\cos\theta - \cos\theta')$$

This work done is stored as potential energy in bar magnet at an angle $\theta$ (when it is rotated from $\theta'$ to $\theta$) and it can be written as

$$U = -p_m B(\cos\theta - \cos\theta') \qquad (3.26)$$

In fact, the equation (3.26) gives the difference in potential energy between the angular positions $\theta'$ and $\theta$. If we choose the reference point as $\theta' = 90^o$, so that second term in the equation becomes zero, the equation (3.26) can be written as

$$U = -p_m B (\cos\theta) \qquad (3.27)$$

The potential energy stored in a bar magnet in a uniform magnetic field is given by

$$U = -\vec{p}_m \cdot \vec{B} \qquad (3.28)$$

**Case 1**

(i) If $\theta = 0^o$, then

$$U = -p_m B (\cos 0^\circ) = -p_m B$$

(ii) If $\theta = 180^o$, then

$$U = -p_m B (\cos 180^\circ) = p_m B$$

From the above two results, We infer that the potential energy of the bar magnet is minimum when it is aligned along the external magnetic field and maximum when the bar magnet is aligned anti-parallel to external magnetic field.

## EXAMPLE 3.7

Consider a magnetic dipole which on switching ON external magnetic field orient only in two possible ways i.e., one along the direction of the magnetic field (parallel to the field) and another anti-parallel to magnetic field. Compute the energy for the possible orientation.

### *Solution*

Let $\vec{p}_m$ be the dipole and before switching ON the external magnetic field, there is no orientation. Therefore, the energy U = 0.

As soon as external magnetic field is switched ON, the magnetic dipole orient parallel ($\theta = 0^o$) to the magnetic field with energy,

$$U_{parallel} = U_{minimum} = -p_m B \cos 0$$

$$U_{parallel} = -p_m B$$

since $\cos 0^o = 1$

Otherwise, the magnetic dipole orients anti-parallel ($\theta = 180^o$) to the magnetic field with energy,

$$U_{anti-parallel} = U_{maximum} = -p_m B \cos 180$$

$$\Rightarrow U_{anti-parallel} = p_m B$$

since $\cos 180^o = -1$



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